Evaluating Text Classification Pipelines: A Comparison of Vectorization, Models, and Interpretability

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Motivation

Text classification is widely used in real-world applications, but optimizing performance and interpretability often involves trade-offs. This project investigates how different modeling decisions affect classification results and explainability, guided by the following key goals:

- **How do different pipeline configurations affect predictive performance?**
  Specifically, how do combinations of vectorization methods (TF-IDF, SBERT, Doc2Vec) and classifiers (Random Forest, MLP, SVM) compare in terms of predictive Performance?
- **How does summarization of the input text impact predictive performance of the pipeline?**
  Does summarizing input texts — either extractively or via LLMs — maintain predictive quality while improving efficiency, particularly in real-time or resource-limited settings?
- **How robust and trustworthy are model explanations?**
  How do LIME explanations compare to those generated by a large language model (LLM) in terms of Robustness? What happens to predictions and explanations when small changes are made to the input?

Dataset & Preproccessing

Two well-established datasets were used to benchmark predictive performance: **20 Newsgroups** and **AG News**. Both contain labeled text samples across multiple topic categories classes allowing evaluation of model performance across different domain complexities.

- 20 Newsgroups: ~18,000 newsgroup posts across **20** categories (e.g. politics, tech, religion)
- AG News: ~120,000 news headlines and descriptions across **4** categories (World, Sports, Business, Sci/Tech)

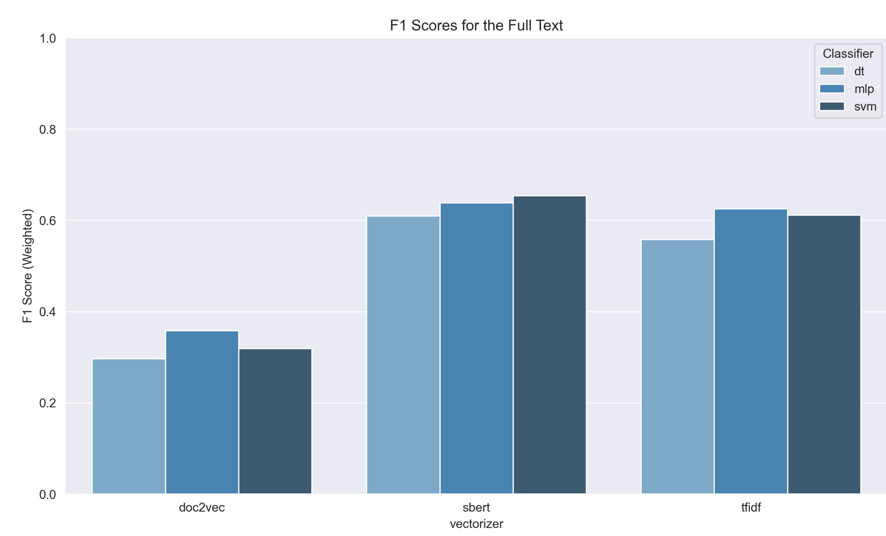
To ensure unbiased learning, header/footer metadata was removed in 20 Newsgroups and any tokens that match class names were also removed to prevent label leakage. These steps address potential shortcuts the models could exploit, particularly important for the explainability component of the study.

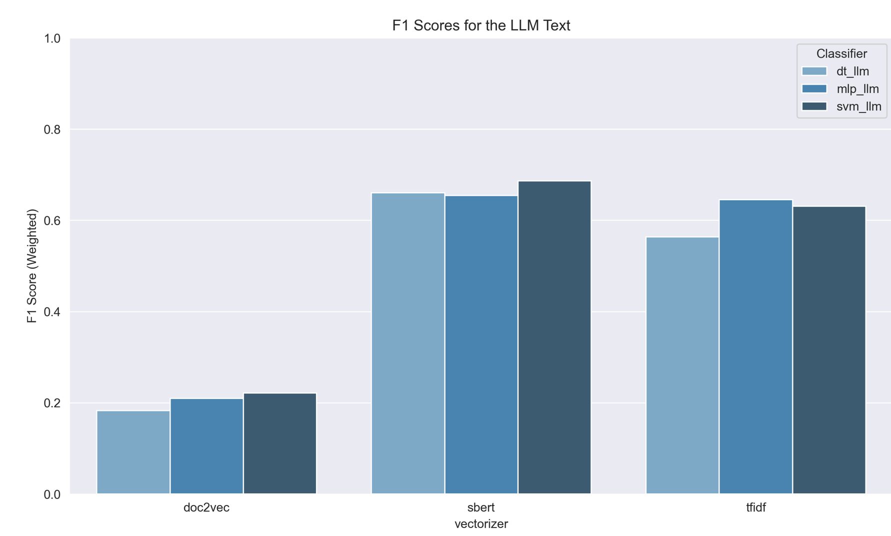
20 Newsgroups

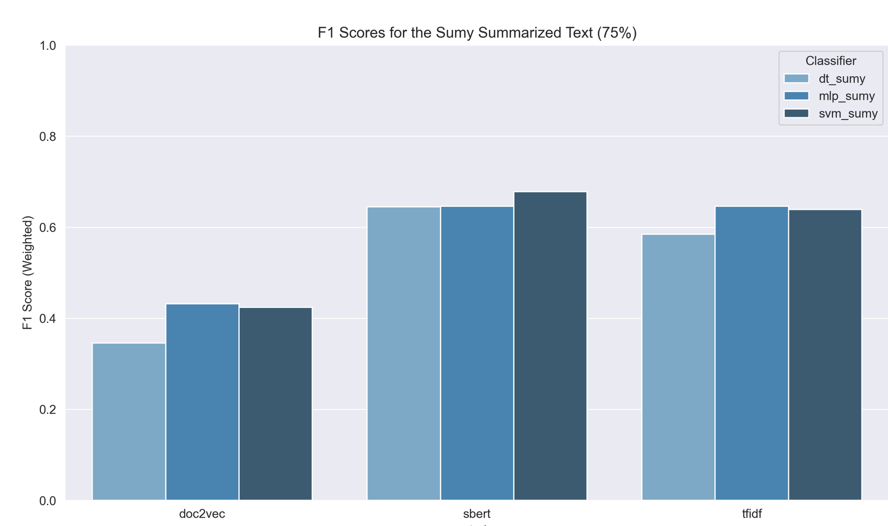
The following graphs show insight into how classification performance on the 20 Newsgroup Dataset is impacted by three different inputs:

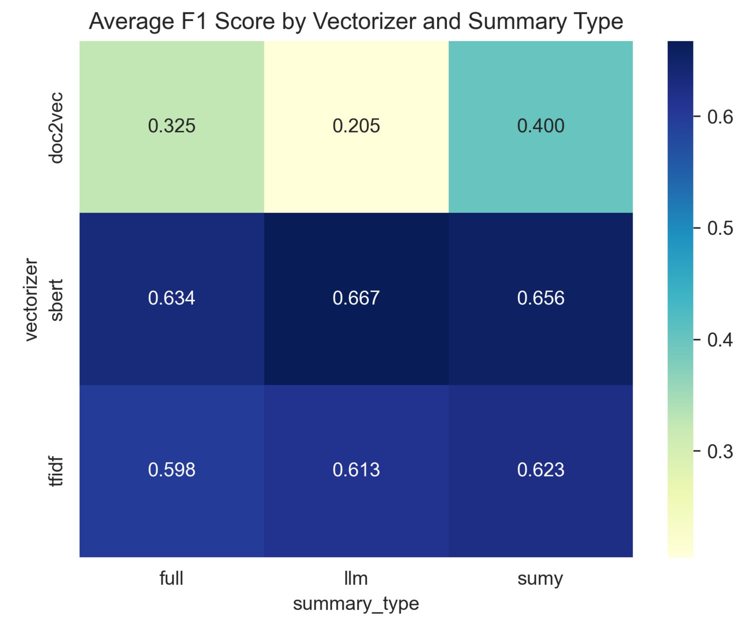
- full texts (~11,000 instances)
- **extractive** summaries generated via Sumy TextRank (~8,800 instances)
- **generative** summaries produced by a large language model (~4,000 instances)

The extractive approach retains original sentences, applied only to texts exceeding six sentences, and compresses each document to 75% of its original length. In the generative approach the local LLM (*llama3.2*) is prompted to summarize the text in 3-4 sentences.









AG News

To benchmark the performance of the pipelines they where retrained and evaluated on the AG News dataset. Taking indication from results of performance on 20 Newsgroup Doc2Vec vectorization technique was disregarded.

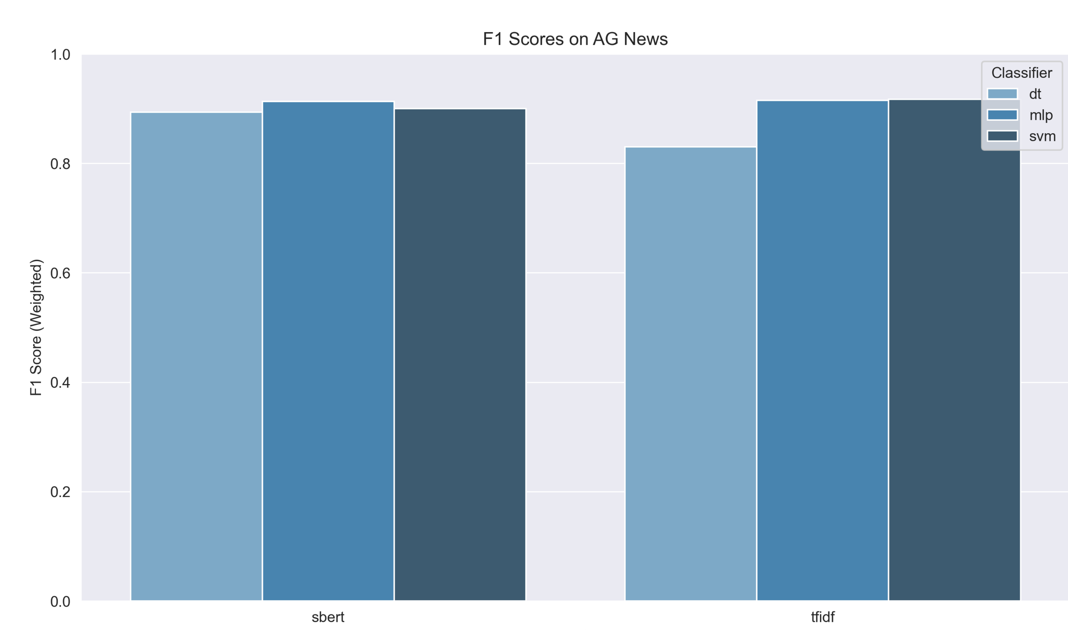
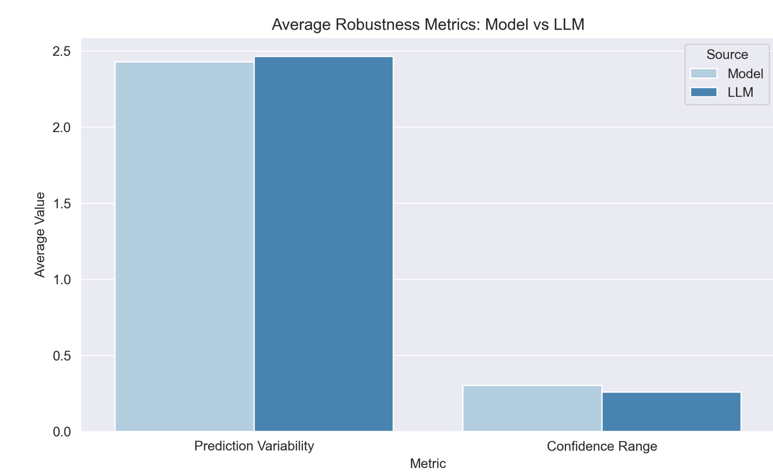


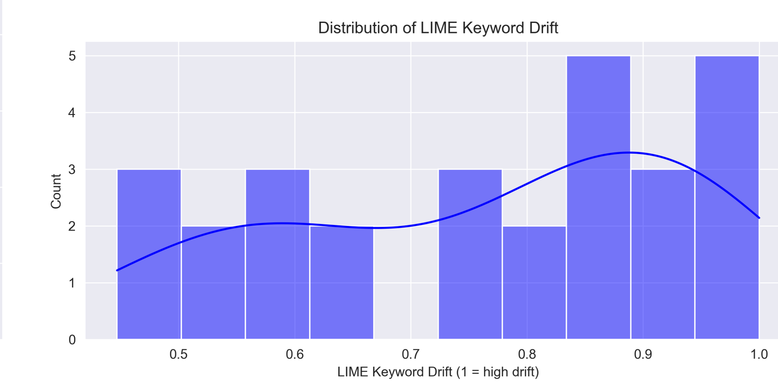
Fig. 1: F1 Scores of pipelines on AG News Dataset

XAI and Reasoning

To evaluate model and explanation robustness, a small subset of **28 base texts** were analyzed. The following steps were performed:

1. **Variation Generation:** For each base text, a local LLM (*llama 3.2*) generated five semantically equivalent text variations.
2. **Prediction and Explanation:** Each variation was classified using both the model and LLM, recording the predicted label, confidence score, and LIME-based explanation (top contributing keywords).
3. **Robustness Scoring:** Each set of 5 variations was scored based on:
   - Number of unique predicted labels (*prediction variability*)
   - Difference between max and min confidence (*confidence drift*)
   - 1 minus the average Jaccard similarity of LIME keywords (*explanation drift*)





Evaluation

The choice of **classifier** seems to have the least overall impact on performance. While decision trees consistently underperformed both support vector machines (SVM) and multilayer perceptrons (MLP) achieved similar and stable results across different combinations. In contrast, the **vectorization method** played a more significant role. Doc2Vec, in particular, yielded significantly poorer results often underperforming regardless of classifier or summarization method.

Summarization level had a moderate but nuanced influence. On average, using LLM-generated summaries resulted in the highest F1 scores, though the improvements were marginal. However, Doc2Vec, despite being the weakest vectorizer overall, showed a remarkable boost in performance when paired with extractive summaries compared to when LLM summaries were used.

Predictions from both the model and the LLM showed limited **robustness** in terms of predictive performance and explanation consistency. Around, ~2.5 unique labels were predicted across just 5 paraphrased variations of the same input, indicating high prediction variability. This was further supported by fluctuating confidence ranges and noticeable drift in LIME explanations.

Conclusion & Limitations

This study examined how vectorization method, classifier choice, and text summarization level affect text classification performance and robustness.

Key findings: Vectorization had the strongest impact on performance, with *SBERT* and *TF-IDF* clearly outperforming *Doc2Vec*. Classifier choice mattered less, with *SVM* and *MLP* performing similarly and better than decision trees. LLM-generated summaries gave slightly better results overall, but extractive summaries surprisingly improved performance for Doc2Vec.

Robustness analysis revealed that both the model and the LLM showed **low stability** across paraphrased inputs — on average, 2.5 unique labels were predicted across just 5 text variations, indicating high sensitivity in both prediction and explanation.

Limitations: Dataset sizes varied across summary types (4K LLM, 8K extractive, full full-text set), limiting direct comparability. Runtime was constrained by the use of a local LLM for summarization, prediction, and variation generation. Additionally, no hyperparameter optimization was performed, as the focus remained on broad comparative trends.